

lemma__EF__lowerbound__aggregation

SCX

2026

1 Lemma E & Lemma F: Second-Order Lower Bound and Multi-State Aggregation

Status: Complete proofs. **Prerequisites:** Lemma A (Bahadur-Rao for Bernoulli), Lemma B (F1 asymptotic expansion), Lemma C (Chernoff information), Lemma D (adaptive threshold). **References:** Bahadur & Rao (1960), Le Cam (1986), van der Vaart (1998), Ingster & Suslina (2003).

1.1 Lemma E: Second-Order Asymptotic Lower Bound (Exact Constant Minimax)

1.1.1 Statement

Let $e_1, \dots, e_M$ be i.i.d. $\text{Bern}(p)$ under either hypothesis:

$$H_0 : p = p_0, \quad H_1 : p = p_1,$$

with $0 \leq p_0 < p_1 \leq 1$ and $(p_0, p_1) \neq (0, 1)$ to avoid degeneracy. Define the Chernoff information

$$\kappa = C(\text{Bern}(p_0), \text{Bern}(p_1)) = \text{KL}(\theta^* \| p_0) = \text{KL}(\theta^* \| p_1),$$

where $\theta^* \in (p_0, p_1)$ is the unique solution of $\text{KL}(\theta \| p_0) = \text{KL}(\theta \| p_1)$. Let

$$\lambda_0^* = \log \frac{\theta^*(1-p_0)}{p_0(1-\theta^*)} > 0, \quad \lambda_1^* = \log \frac{\theta^*(1-p_1)}{p_1(1-\theta^*)} < 0,$$

and set $D = \lambda_0^* + |\lambda_1^*|$, $s = |\lambda_1^*|/D$.

For any noise detection algorithm $\mathcal{A}$ (a measurable map $\{0, 1\}^M \rightarrow \{0, 1\}$) let $\text{F1}_{\mathcal{A}}(M)$ denote its expected F_1 score on a sample where $\Pr(\text{noise}) = \eta \in (0, 1)$.

Theorem (Lower Bound). For any such algorithm $\mathcal{A}$,

$$\liminf_{M \rightarrow \infty} e^{M\kappa} \sqrt{2\pi M} (1 - \text{F1}_{\mathcal{A}}(M)) \geq \frac{C_{\min}}{\eta},$$

where

$$C_{\min} = \frac{\eta}{2} \left(\frac{1-\eta}{\eta} \right)^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}$$

Equivalently, writing $r = \lambda_0^*/D = 1 - s$,

$$C_{\min} = \frac{\eta^r (1-\eta)^{1-r}}{2} \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1-\theta^*)}}.$$

The constant is sharp: the Bayes-optimal threshold test (Lemma D) attains the limit with this same constant.

1.1.2 Proof

The proof proceeds in five parts.

Part 1: Reduction to a binary hypothesis test Within a fixed state s , Assumption (A5) guarantees that the expert errors are conditionally i.i.d. given the state. Under the clean label distribution the error probability is $p_0 = \mu_s$; under the noisy label distribution it is $p_1 = 1 - C_{\text{bal}} \cdot \mu_s / (K - 1)$. Hence the algorithm's input $(e_1, \dots, e_M)$ constitutes M i.i.d. draws from $\text{Bern}(p_0)$ under H_0 and from $\text{Bern}(p_1)$ under H_1 .

Let the algorithm's decision be $\delta_M \in \{0, 1\}$, where $\delta_M = 1$ means “flag as noisy.” The Type I and Type II errors are

$$\alpha_M = \Pr(\delta_M = 1 \mid H_0), \quad \beta_M = \Pr(\delta_M = 0 \mid H_1).$$

From Lemma B, the F_1 score satisfies

$$1 - F_1 = \frac{1 - \eta}{2\eta} \alpha_M + \frac{1}{2} \beta_M + o(1),$$

where the $o(1)$ term is uniform over all decision rules and decays as $O(e^{-2M\kappa})$ (negligible for the leading constant). Define the weighted risk

$$R_M(\delta_M) = w_0 \alpha_M + w_1 \beta_M, \quad w_0 = \frac{1 - \eta}{2\eta}, \quad w_1 = \frac{1}{2}.$$

Our task is therefore to lower-bound $\liminf e^{M\kappa} \sqrt{2\pi M} R_M(\delta_M)$ over all measurable decision rules δ_M .

Part 2: Application of the Neyman-Pearson lemma Consider the simple-versus-simple testing problem $H_0 : p = p_0$ vs $H_1 : p = p_1$. The likelihood ratio for M observations is

$$L_M(e) = \prod_{m=1}^M \frac{p_1^{e_m} (1 - p_1)^{1-e_m}}{p_0^{e_m} (1 - p_0)^{1-e_m}}.$$

Because $p_1 > p_0$, L_M is strictly increasing in $C_M = M^{-1} \sum e_m$; consequently, the most powerful level- α test is a threshold test on C_M .

Fix any decision rule δ_M and let $\alpha = \alpha(\delta_M)$. The Neyman-Pearson lemma (Lehmann & Romano, Thm 3.2.1) states that among all tests with Type I error $\leq \alpha$, the likelihood ratio test (equivalently, the threshold test on C_M) minimizes the Type II error. Hence

$$\beta(\delta_M) \geq \beta_{\text{NP}}(\alpha),$$

where $\beta_{\text{NP}}(\alpha)$ is the Type II error of the most powerful level- α test. Consequently

$$R_M(\delta_M) = w_0\alpha + w_1\beta(\delta_M) \geq w_0\alpha + w_1\beta_{\text{NP}}(\alpha) \geq \min_{\alpha \in [0,1]} \{w_0\alpha + w_1\beta_{\text{NP}}(\alpha)\}.$$

The function $f(\alpha) = w_0\alpha + w_1\beta_{\text{NP}}(\alpha)$ is convex (the ROC curve $\alpha \mapsto \beta_{\text{NP}}(\alpha)$ is concave). Its minimizer α^* satisfies $f'(\alpha^*) = 0$, i.e.

$$w_0 + w_1 \beta'_{\text{NP}}(\alpha^*) = 0.$$

For the Neyman-Pearson test, the slope of the ROC curve at α^* equals $-1/\tau^*$, where τ^* is the likelihood-ratio threshold. This gives $\tau^* = w_0/w_1 = (1 - \eta)/\eta$; the minimizer is exactly the Bayes decision rule with prior odds $w_0 : w_1$ against H_0 .

Therefore, for **any** algorithm $\mathcal{A}$, the weighted risk is at least the risk of the Bayes test:

$$R_M(\delta_M) \geq R_M^* := w_0 \alpha_M^* + w_1 \beta_M^*,$$

where

$$\delta_M^*(e) = \mathbf{1}\{L_M(e) > (1 - \eta)/\eta\}$$

(with arbitrary tie-breaking, which affects the risk by $O(e^{-M\kappa}/M)$ and is irrelevant asymptotically).

Part 3: Exact asymptotics of the Bayes test The Bayes test rejects H_0 when the log-likelihood ratio exceeds $\log((1 - \eta)/\eta)$. Write

$$\Delta = \log \frac{p_1(1 - p_0)}{p_0(1 - p_1)} > 0, \quad \Delta_0 = \log \frac{1 - p_1}{1 - p_0}.$$

Then $\log L_M = M(C_M \Delta + \Delta_0)$, and the rejection region is

$$C_M > -\frac{\Delta_0}{\Delta} + \frac{1}{M\Delta} \log \frac{1 - \eta}{\eta}.$$

The constant $-\Delta_0/\Delta$ is precisely the Chernoff point θ^* , since $\text{KL}(\theta^* \| p_0) = \text{KL}(\theta^* \| p_1)$ implies $\theta^* \Delta + \Delta_0 = 0$. Hence the threshold is

$$t_M = \theta^* + \frac{1}{M\Delta} \log \frac{1 - \eta}{\eta}.$$

A key identity (proved in Lemma C) relates Δ to the derivatives of the KL divergence at θ^* :

$$\Delta = \lambda_0^* - \lambda_1^* = \lambda_0^* + |\lambda_1^*| = D.$$

Thus $t_M - \theta^* = \delta_M$ with

$$\delta_M = \frac{1}{MD} \log \frac{1-\eta}{\eta} = O\left(\frac{1}{M}\right).$$

The error probabilities of the Bayes test are

$$\alpha_M^* = \Pr_{p_0}(C_M > t_M), \quad \beta_M^* = \Pr_{p_1}(C_M \leq t_M),$$

up to a $O(e^{-M\kappa}/M)$ correction for possible randomization at the boundary.

Apply the Bahadur-Rao theorem (Lemma A) to each:

$$\alpha_M^* = \frac{\exp(-M \text{KL}(t_M \| p_0))}{\lambda_0^*(t_M) \sqrt{2\pi M t_M(1-t_M)}} (1 + O(M^{-1})),$$

$$\beta_M^* = \frac{\exp(-M \text{KL}(t_M \| p_1))}{|\lambda_1^*(t_M)| \sqrt{2\pi M t_M(1-t_M)}} (1 + O(M^{-1})).$$

Since $t_M \rightarrow \theta^*$, we have $\lambda_0^*(t_M) \rightarrow \lambda_0^*$, $|\lambda_1^*(t_M)| \rightarrow |\lambda_1^*|$, and $t_M(1-t_M) \rightarrow \theta^*(1-\theta^*)$; the $O(M^{-1})$ corrections in the prefactors are uniform in a neighbourhood of θ^* .

Part 4: Expansion around the Chernoff point Write $\delta = \delta_M$ and expand the KL divergences:

$$\text{KL}(\theta^* + \delta \| p_0) = \kappa + \lambda_0^* \delta + \frac{\delta^2}{2\theta^*(1-\theta^*)} + O(\delta^3),$$

$$\text{KL}(\theta^* + \delta \| p_1) = \kappa - |\lambda_1^*| \delta + \frac{\delta^2}{2\theta^*(1-\theta^*)} + O(\delta^3).$$

The quadratic term satisfies $M\delta^2 = O(M^{-1}) \rightarrow 0$ because $\delta = O(M^{-1})$; the cubic term is $O(M^{-2})$. Hence